Where does Qwen3-32B *hold* a paper's authors?

A layer-by-layer readout of author names for 9,108 papers

Stages 10–11 · jlens

Qwen3-32B · 2026-10-27

The question

target (4,621 papers)
can't write the list (R0/R1)
but picks the true list 4/4 times

control (2,220 papers)
can't write the list (R0/R1)
and can't pick the true list

If the model already **knows** target authors in its middle layers and only fails to say them, a lens on those layers should read the right name **better for target than control**.

If the knowledge is only assembled at the end, the gap should appear only near the output.

So: at which layer does target pull ahead of control?

The data

9,108 papers, 8 fields

33,811 author names

149,999 name tokens scored

13 citation bins (0 to 63,901)

Source: the **Dreaming Up Authors** evaluation set. 9,999,863 random OAG v2 records + Semantic Scholar metadata (Sep 2025) -> quality filters -> 1.92 M papers -> up to 100 per field x citation bin

Groups (group lead's labelling run, same model)

target 4,621 · control 2,220

guard (writes the authors) 214 · excluded 2,053

Medicine, biology, CS, psychology, chemistry, maths, physics, materials science: ~1,100 papers each

Name formats

73% start with an initial ("S. Bree")

26% start with a full given name ("Yan Zhao")

How groups and contrasts are built

labels: group lead's pipeline on the same Qwen3-32B · we use them unchanged

open answer -> R0 / R1 / R2

greedy, chat template

fuzzy name match: M, H, U

HR2 = (H+U) / (M+H+U)

R2: HR2 <= 0.5 · R0: M = 0 · R1: M > 0

a fixed rule, no human or model judge

multiple choice -> target / control

true list + 3 real teams of other papers

(same field, team size, citation bin, ~year)

4 orders · p(letter) from next token

target: 4/4 and mean p(true) >= 0.5

control: <= 1/4

5 false titles per paper

same field, own year kept

min-cost on |dyear|/5 + |dauthors|

+ |dtitle tokens| + 1.5 |dcitation bin|

no shared surname, Jaccard <= 0.6

79% strict tier

statistics

target vs control only within citation bin

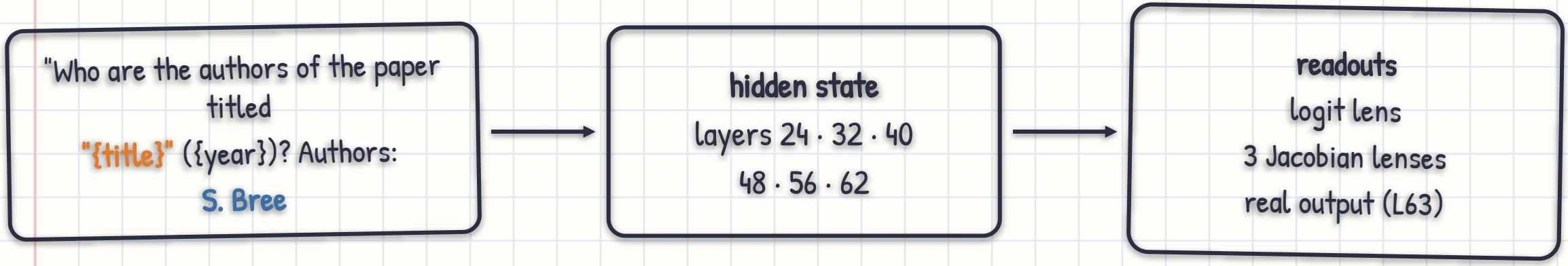
10 bins with >= 30 per group

weight n_t n_c / (n_t + n_c)

95% bootstrap over papers, 2,000

sensitivity: drop bins 0-4

How we read the model



Score = rank of the correct next name token,

as log10(1 / rank): 0 = rank 1, -1 = rank 10

7 title conditions: the true title, 5 matched false titles, no title

title-specific signal

= true title - mean(5 false titles)

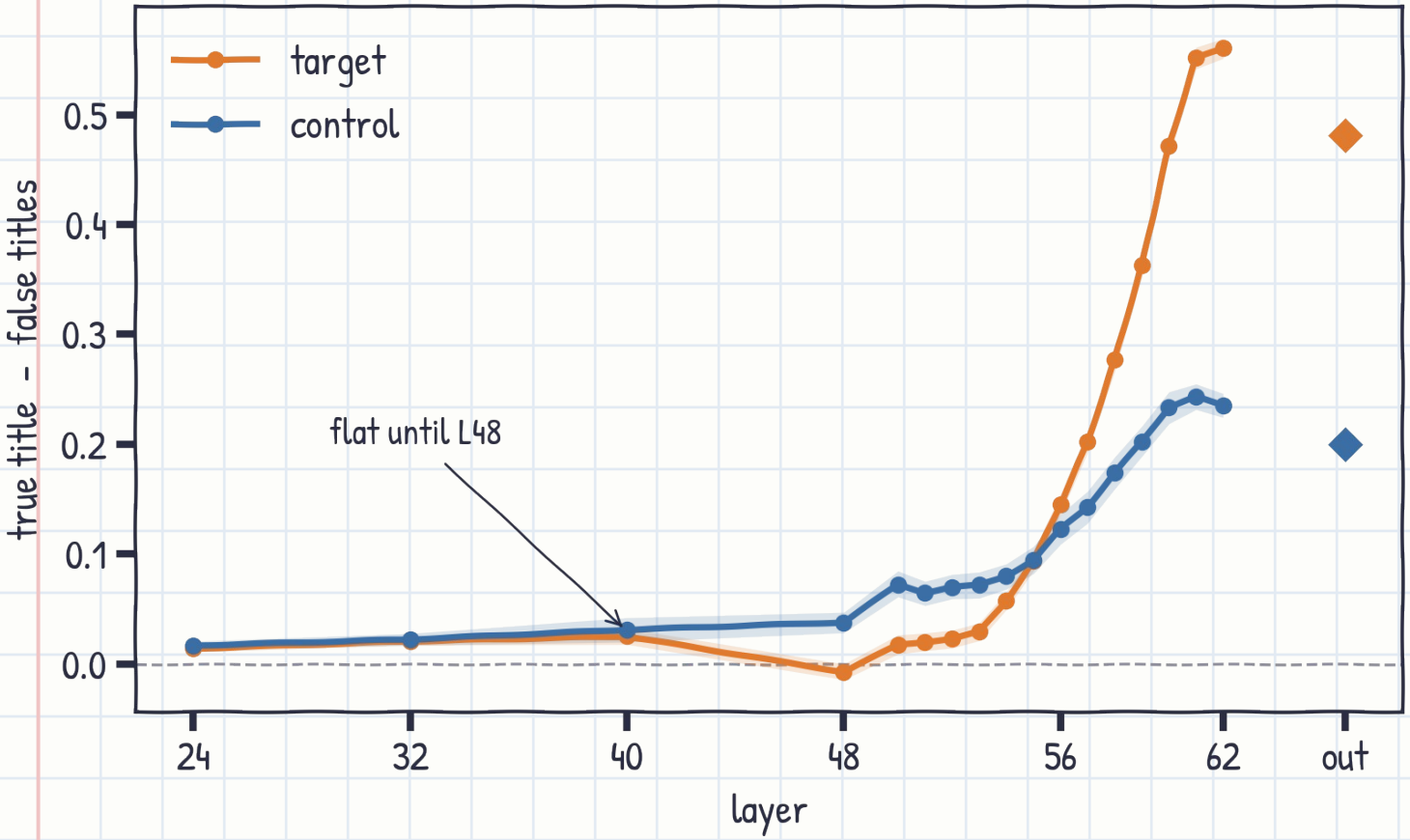
Same name, same tokens: what is left is knowing

this paper

Lenses (model weights never change; Jacobians to L63 averaged over prompts): public = 557 WikiText-103 prompts (Neuronpedia) · self100 = 100 frozen WikiText-103 prompts · fitA = 100 other WikiText-103 prompts (Stage 9: one skipped prompt replaced -> 100 valid)

Knowing the paper shows up only at the end

title-specific signal · public Jacobian lens (lines) and model output (diamonds)



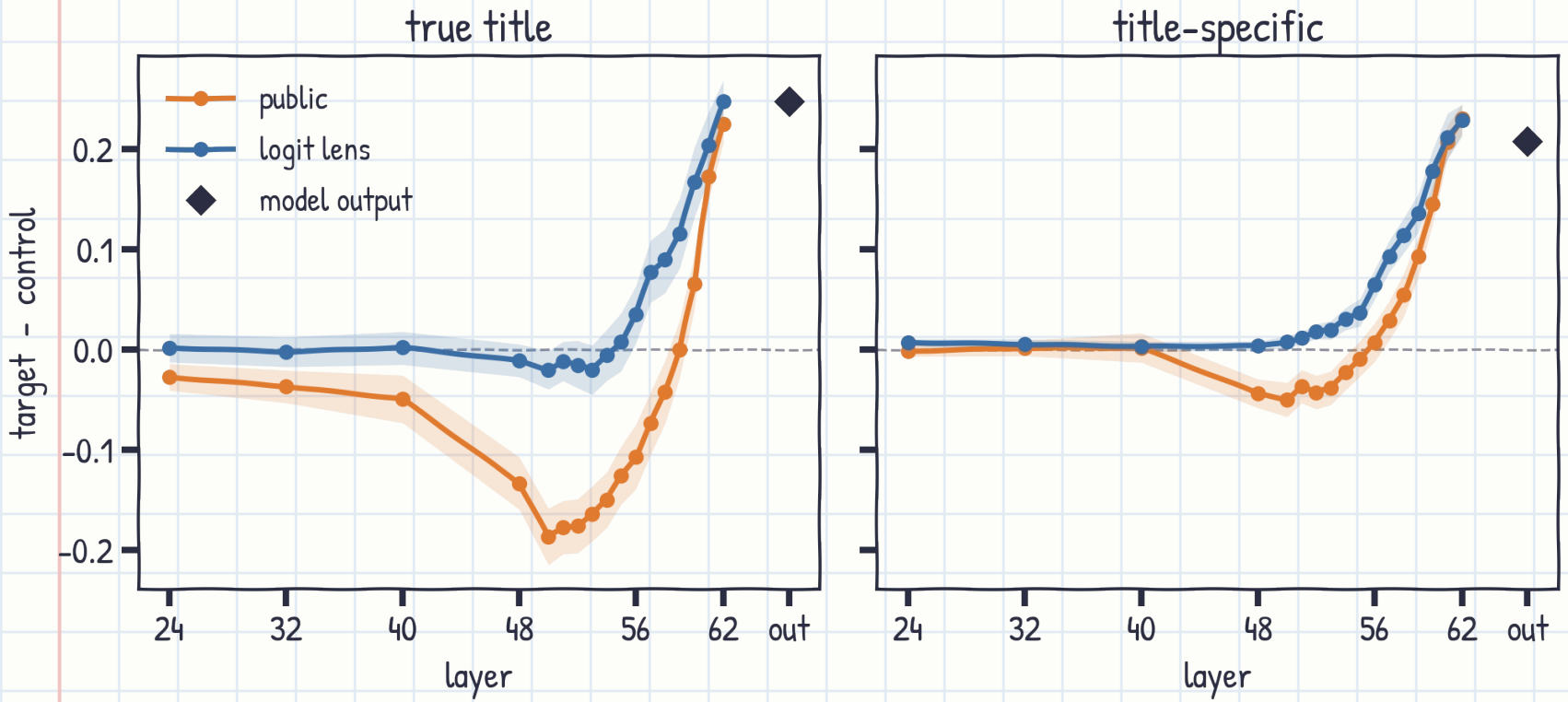
L24–L48: ~0 for both groups

rises at **L56**

L62: target +0.56 vs control +0.24

Target pulls ahead only in the last layers

target − control, within citation bin, 95% CI



Middle layers: **no head start**

L62 / output:

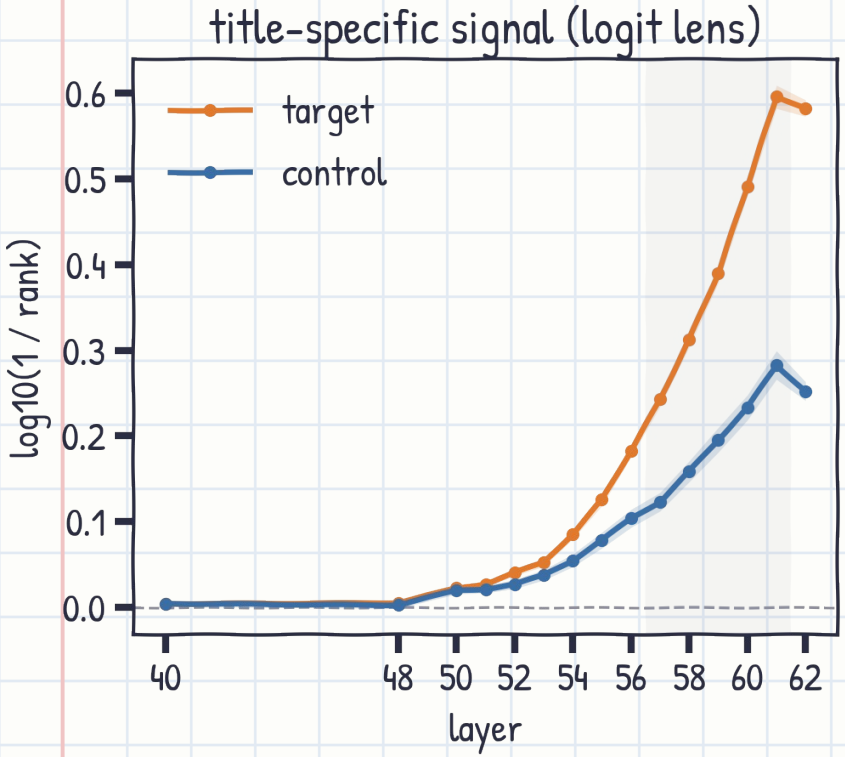
+0.21 to +0.23

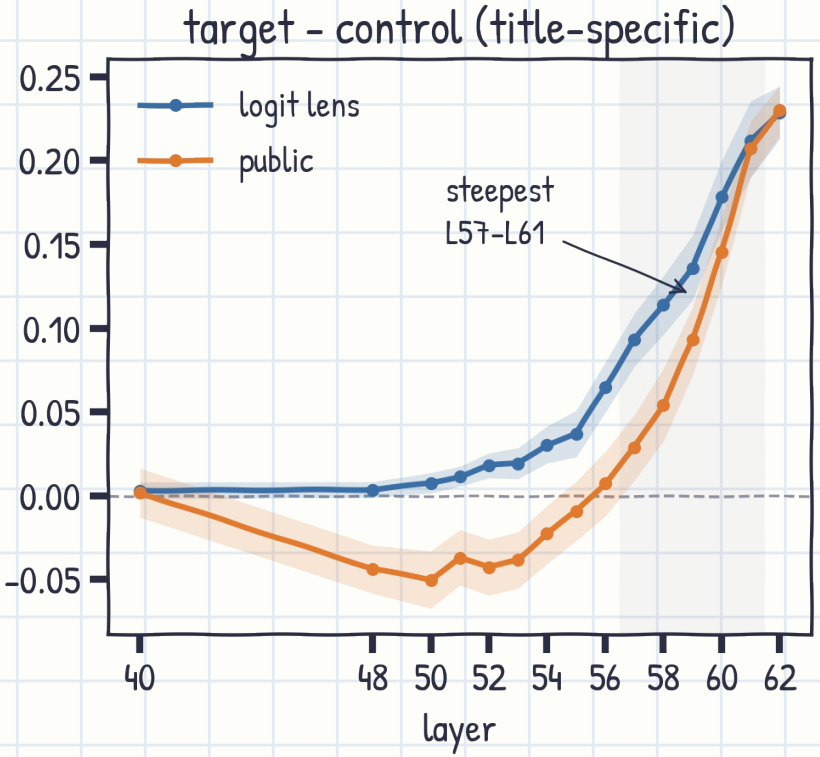
almost all title-specific

R0 only (no author written): output +0.199 vs +0.207

It builds up over the last dozen layers

fine layers L40–L62 · title-specific signal · 95% CI





Logit lens: target - control

+0.007 at L50

+0.23 at L62

steepest rise **L57–L61**

Stage 11: intervene, do not just read

593 + 593 papers: min(60, n_t, n_c) per citation bin, seed 11 · patch uses false_0 · knockout masks all heads · all 9,108 for probe and chain

patch

copy the true-title hidden state into the false-title run at one layer

knock out

block attention to the title tokens (all heads) from layer l to the last layer

probe

linear map from the last title token to the output state

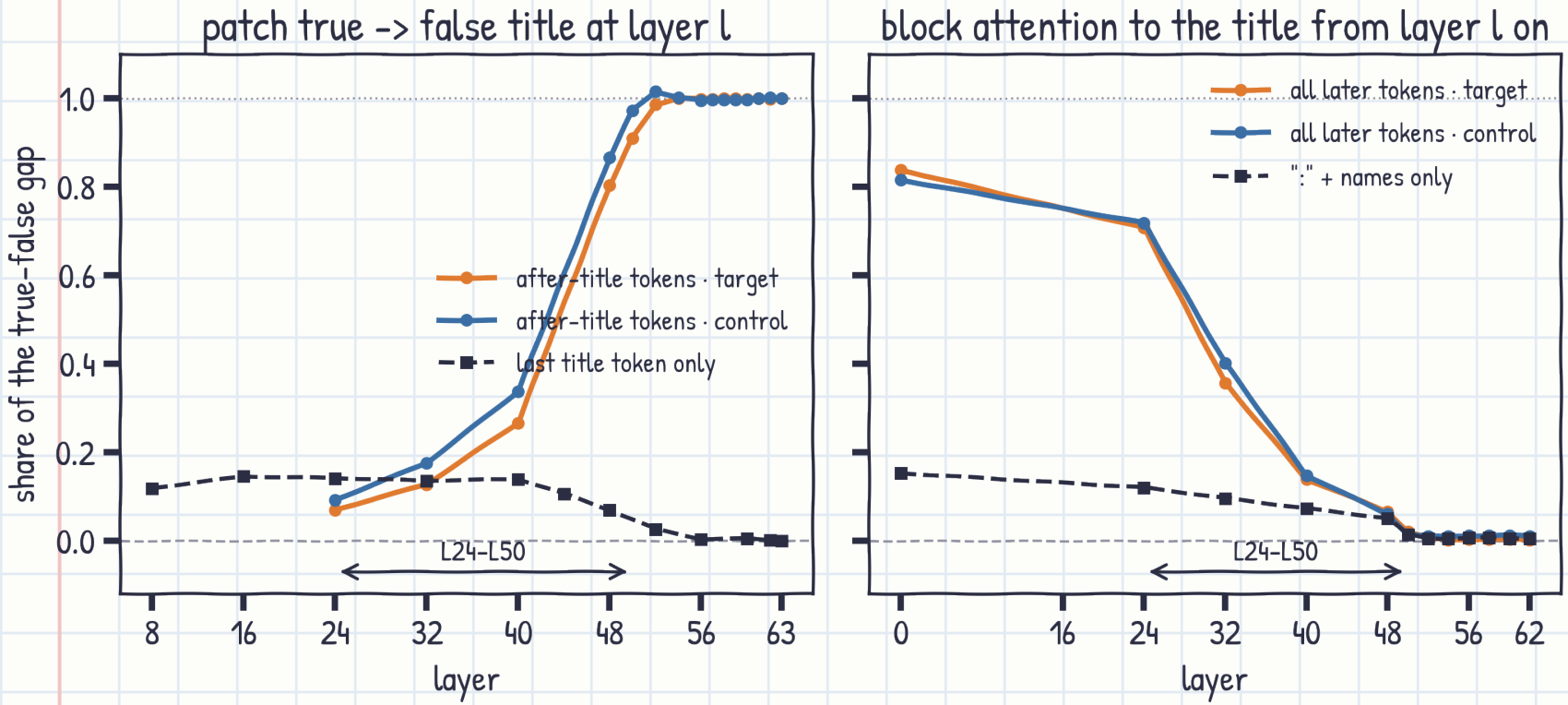
chain

"Authors: A1, A2, ..." instead of each author alone

Score: rank of the correct name token at the model output; "share of the true-false gap" = (intervened − false) / (true − false).

The title is read in L24-L50, early

model output · share of the true-false gap · CIs about +/-0.02



L52+: title no longer needed

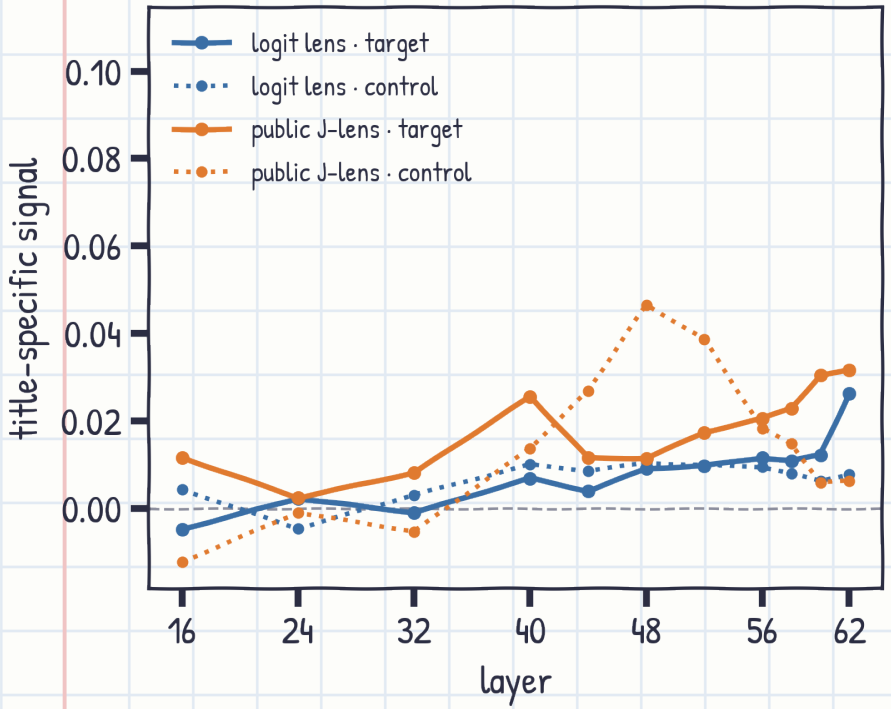
info moved to the tokens **right after** the title

same timing for target and control

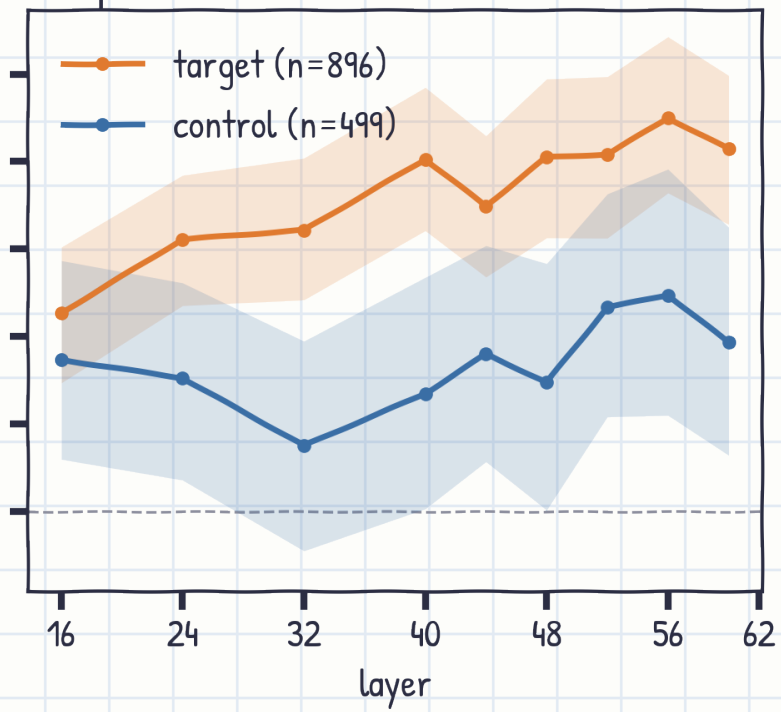
Read early: a probe finds it at the title

title-specific signal at the last title token · lenses: all 9,108 papers · probe: re-checked, title reuse removed: 1,863 papers (896 target, 499 control)

lens readout at the last title token



linear probe at the last title token (re-checked)



lenses: ~0.01–0.03

probe: 0.05 -> 0.08

target > control by +0.04–0.08

re-checked: team folds, title reuse removed

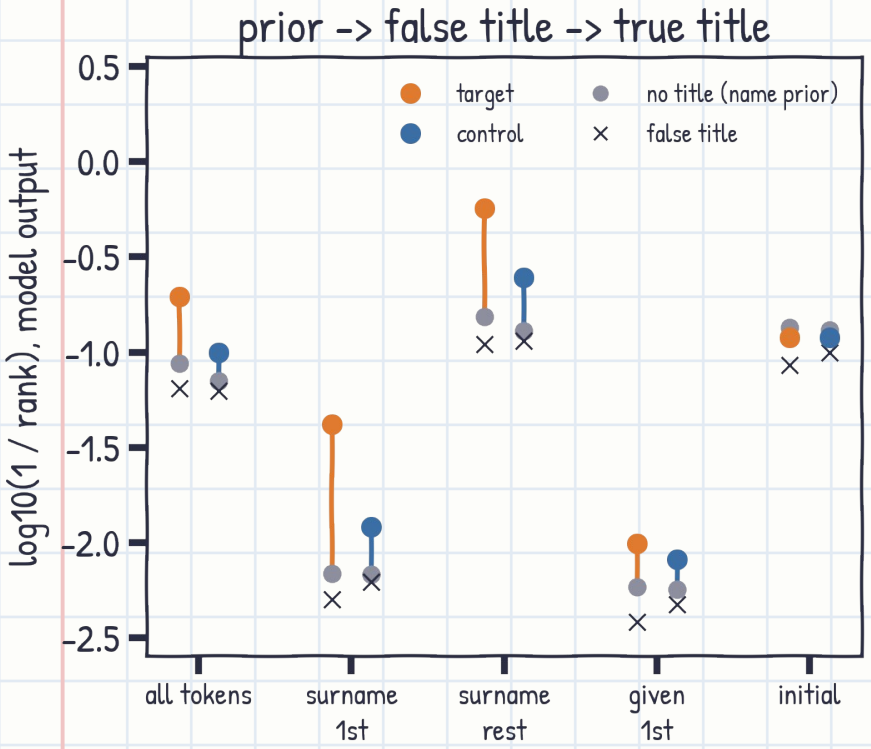
scores the first name token only (mostly an initial); subtracts false_0 only

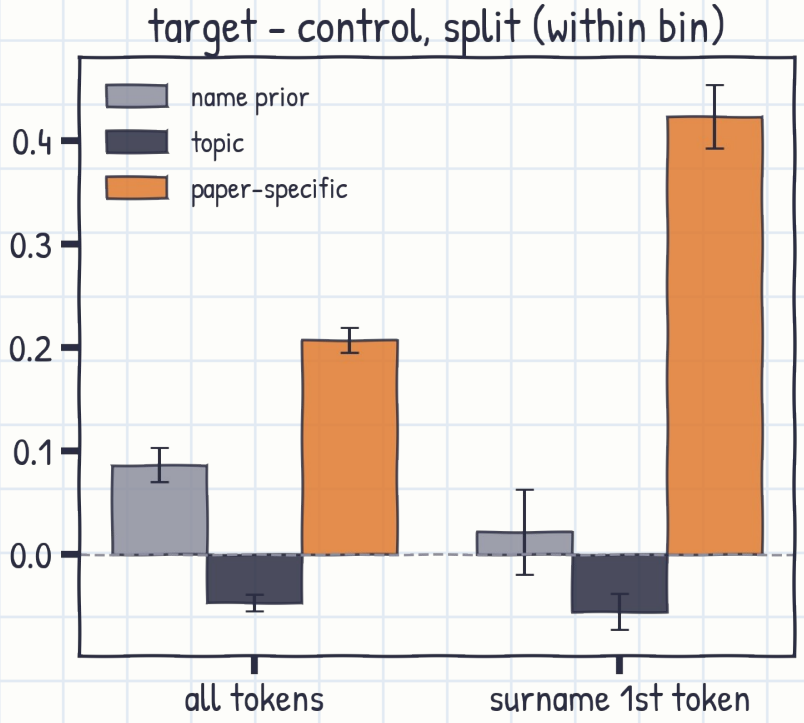
read early (title, from L24), spoken late (names, L57–61)

Stage 11

How much is just the name prior?

model output · no title vs false title vs true title · all 9,108 papers





no title: rank ~12

target - control: +0.09 prior, +0.21 paper

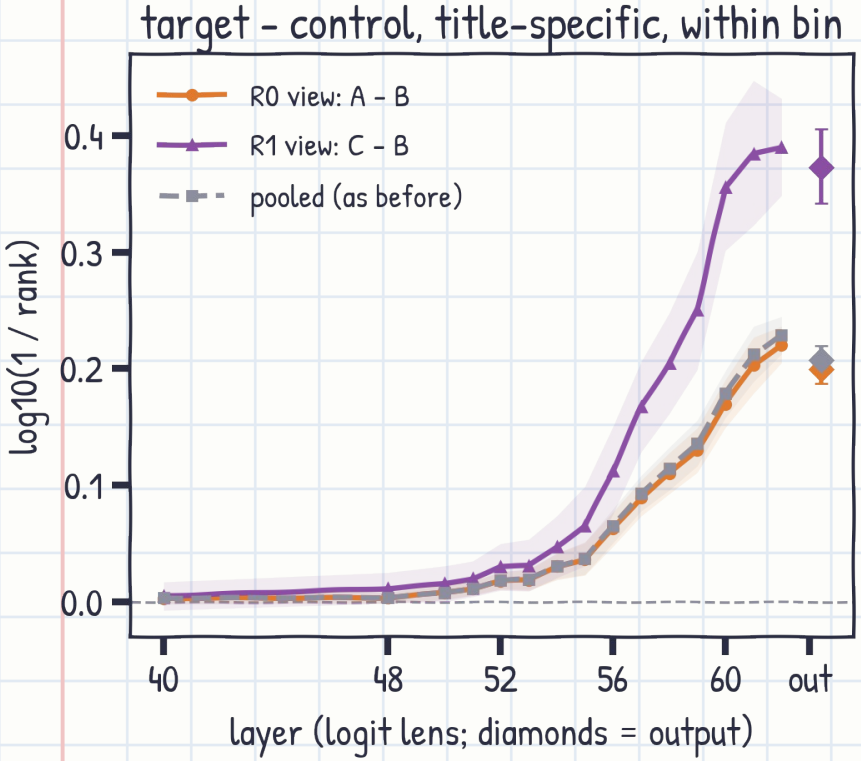
surname 1st: prior ~0, paper +0.42

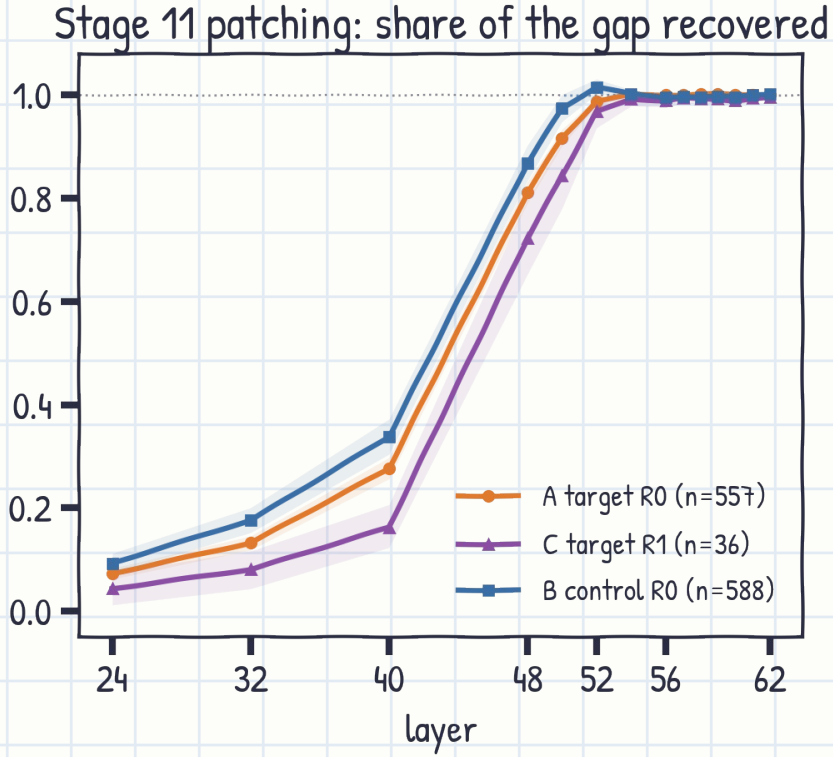
prior vs recognition rho **0.09**

Stage 10–11

Is it just the R1 papers? No.

R0 = no true author written · R1 = a few written · A/B/C/D = target/control x R0/R1





R0 only (A - B):

output +0.199 vs +0.207, same shape

R1 (C - B): +0.37, also late

C - A within target: +0.11, from L56

patching: R1 a bit later, done by L52

What we learned: read early, spoken late

Read early	Causally the title is read in L24–L50 and gathered at the name positions by L52, then no longer needed. A probe at the title token reads a little of it from L24 (re-checked).
Spoken late	At the name positions, paper-specific author knowledge becomes lens-readable only from ~L50, fastest over L57–L61.
No lens lead	Read with a lens at the names, target is not ahead of control in middle layers; the +0.21 gap is at the end (probe at the title: target − control +0.04–0.08, re-checked).
Real knowledge	Name prior is ~1/3 of the gap overall but ~0 on surnames; the rest is paper-specific and grows with citations.
Lenses agree	Jacobian lenses read more in middle layers, but nothing target-specific.
Name format	Initials carry little; full given names carry as much as surnames and help predict them.

Caveats and next steps

Stage 12 plan

Caveats

- Readout is correlational; patching/knockout fix the timing, not the mechanism inside L24–L50
- Parity tolerance raised 0.25 -> 0.5 after the fact
- Groups are model-specific; low-citation "recognition" is near chance
- Few control papers above 1,000 citations
- Label run settings (max length, thinking mode) not in the package

Stage 12 (running)

1. Patch single positions (quote, year, "Authors:") – done, analysing
2. Attention vs MLP patching in L8–L56
3. A lens fitted on author prompts, read at L24–L62
4. Name-prior baselines: distractor lists without a title; a prompt with no paper

Appendix

checks · readout levels · author chain · probe controls · lenses · citations · name formats

Can we trust the numbers?

fast = slow

One forward pass vs one per prefix: max gap 0.37 logit (bf16 rounding); ranks 1-41 identical

matches Stage 6

972 tokens: same token ids, median log-prob gap ~1e-5

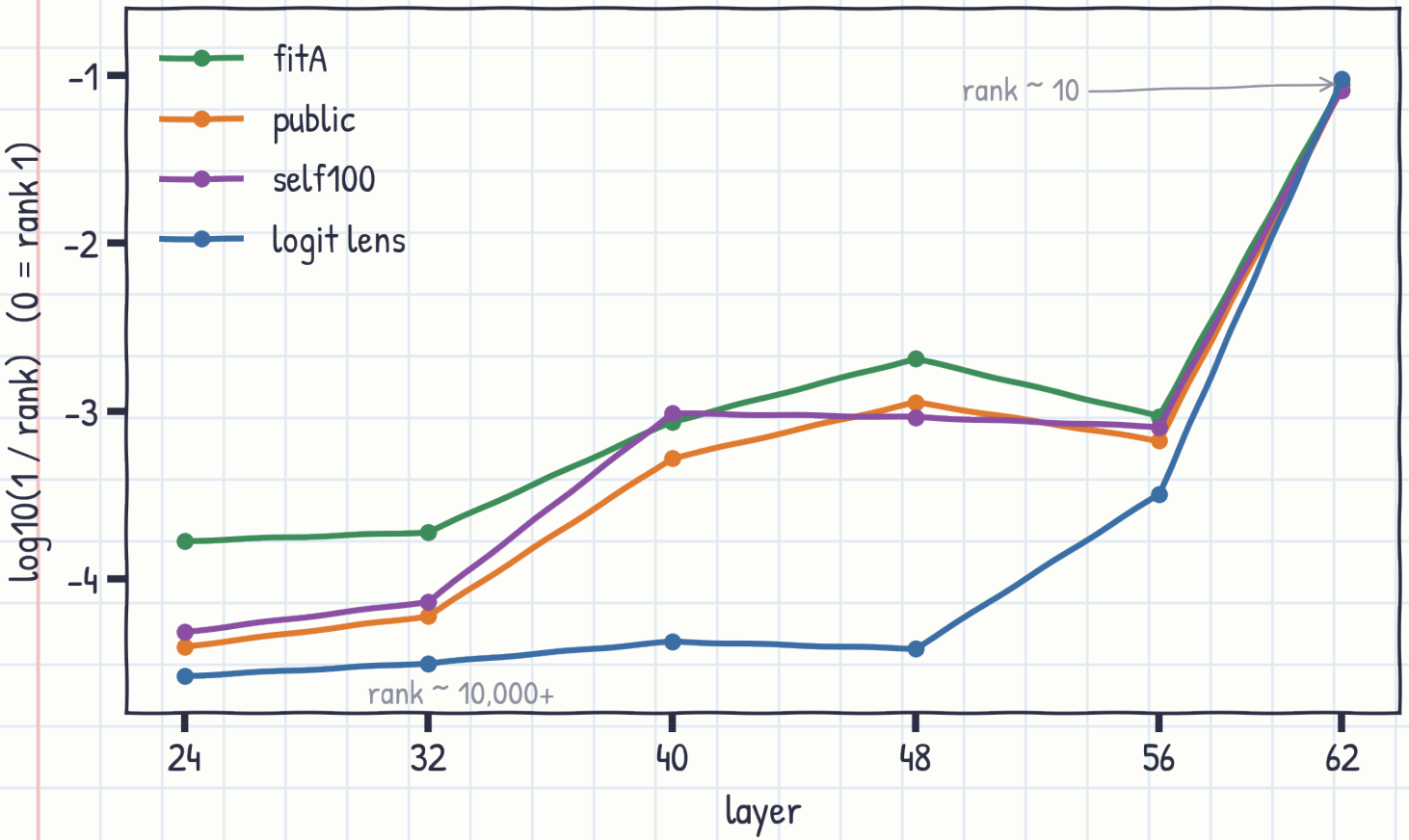
negative control

Punctuation tokens show ~0 title-specific signal at every layer

Caveat: the parity tolerance was raised 0.25 -> 0.5 after seeing these bf16 gaps.

Names become readable late

all 9,108 papers · true title



Logit lens: almost nothing until L48

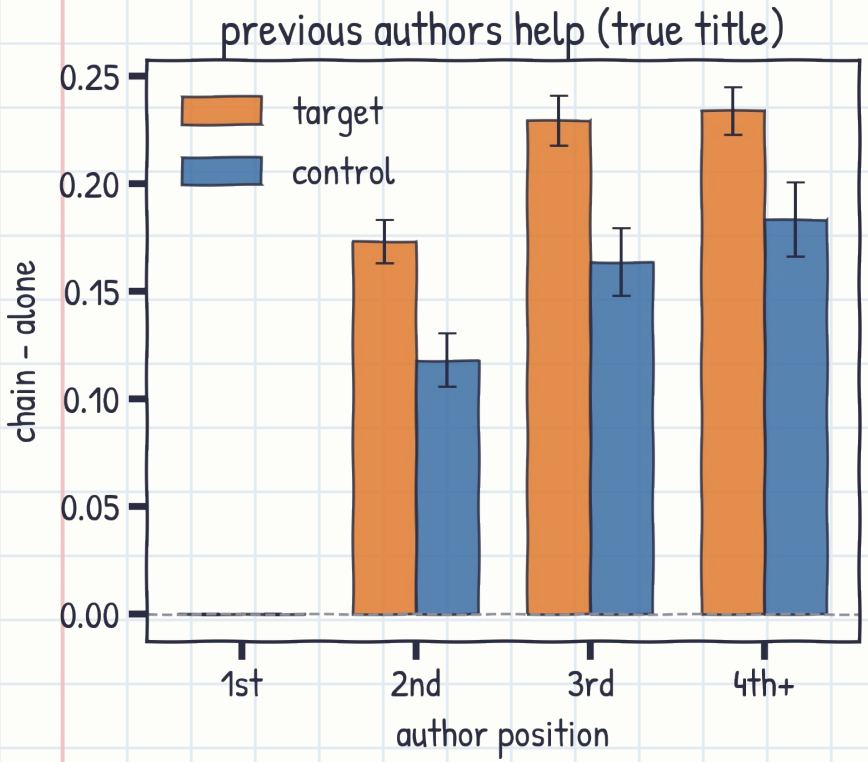
Jacobian lenses read more from L40

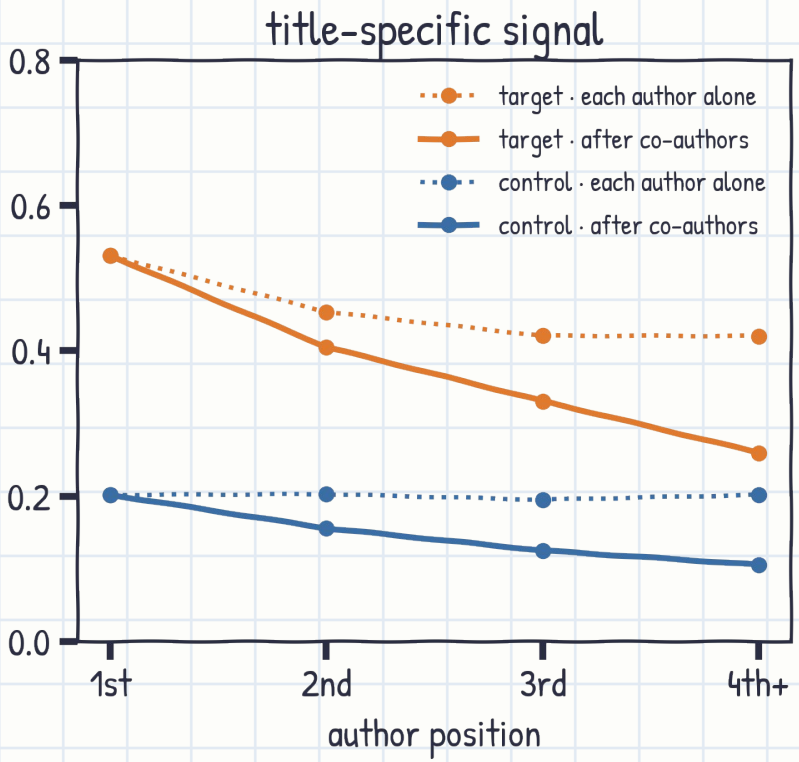
Everything jumps at L62 (rank ~10)

Stage 11

Co-authors help, and partly replace the title

model output · all 9,108 papers · 95% CI





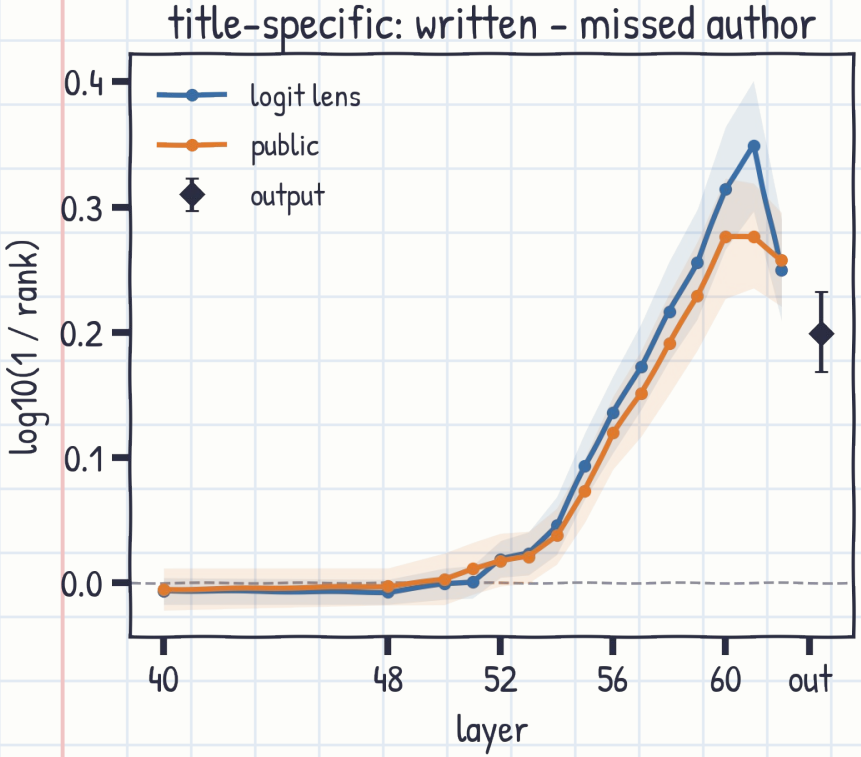
later authors +0.17 to +0.23

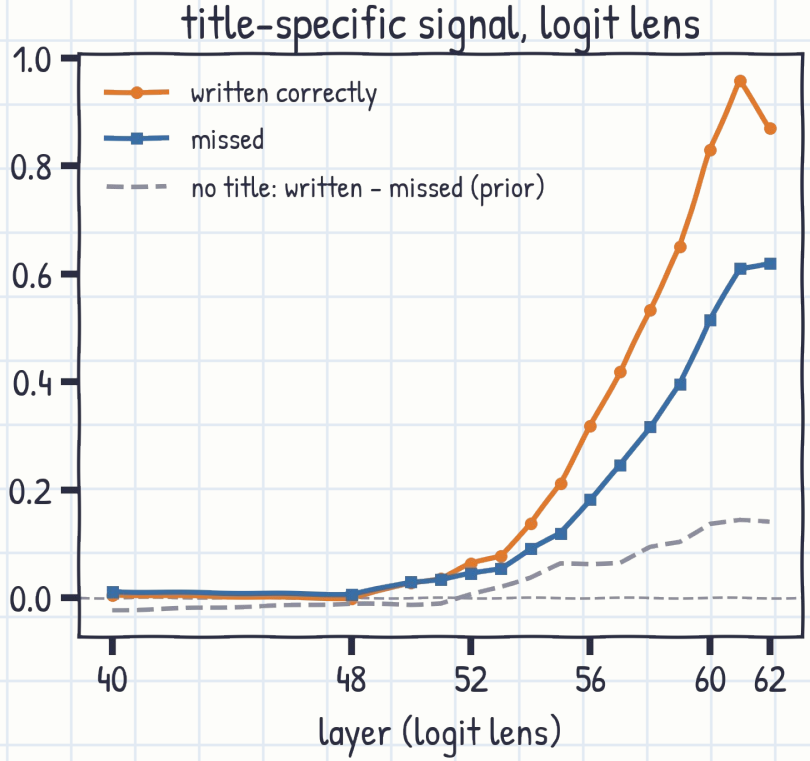
title-specific part **shrinks**

target > control still

Authors it wrote vs authors it missed

inside 495 R1 papers · paired within paper · 95% CI





written − missed, output: +0.20

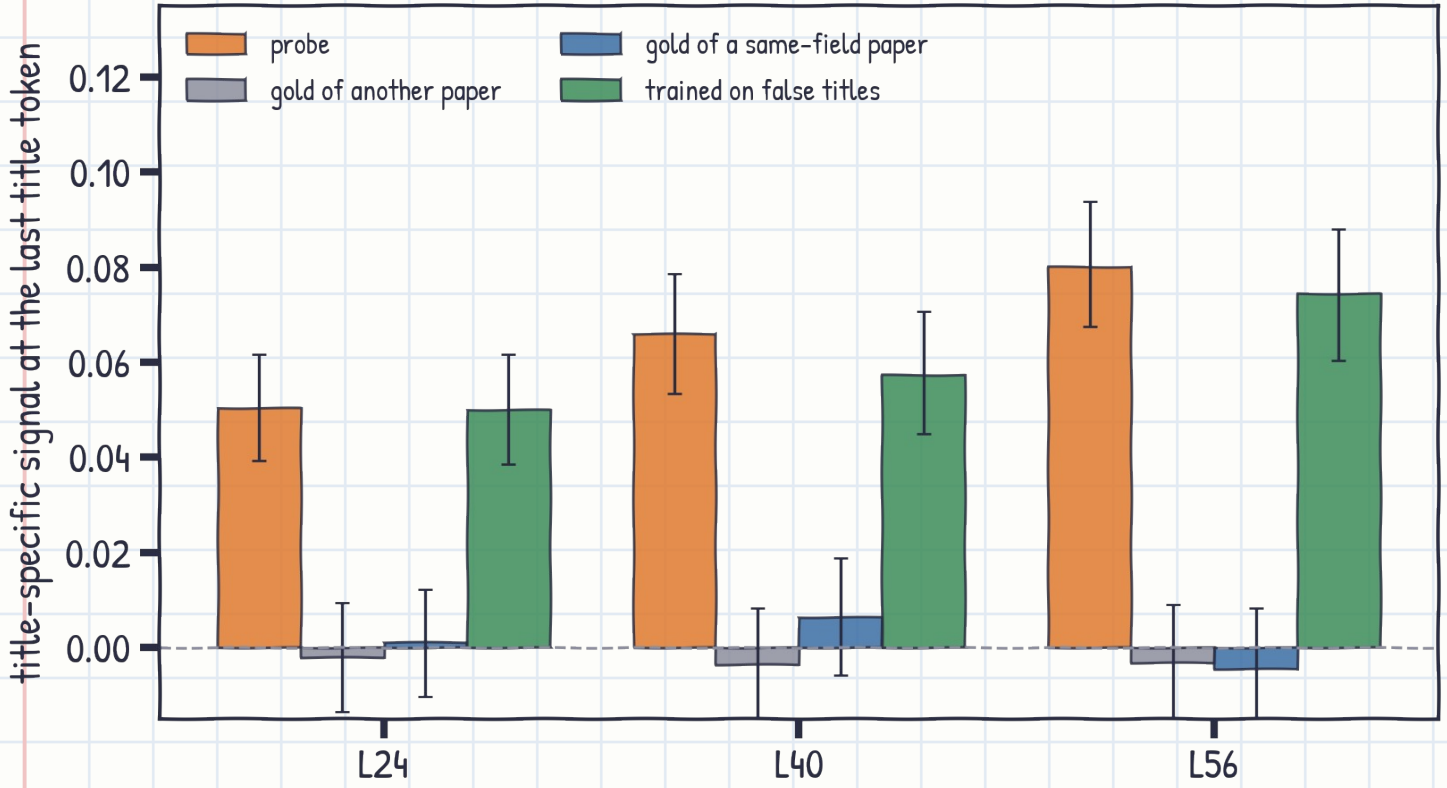
~0 up to L52, grows from **L54**

no title (prior): +0.10

same without first authors

The probe signal is paper-specific

last title token · 5-fold ridge probe · re-checked: author-team folds, title reuse removed



another paper's authors: 0

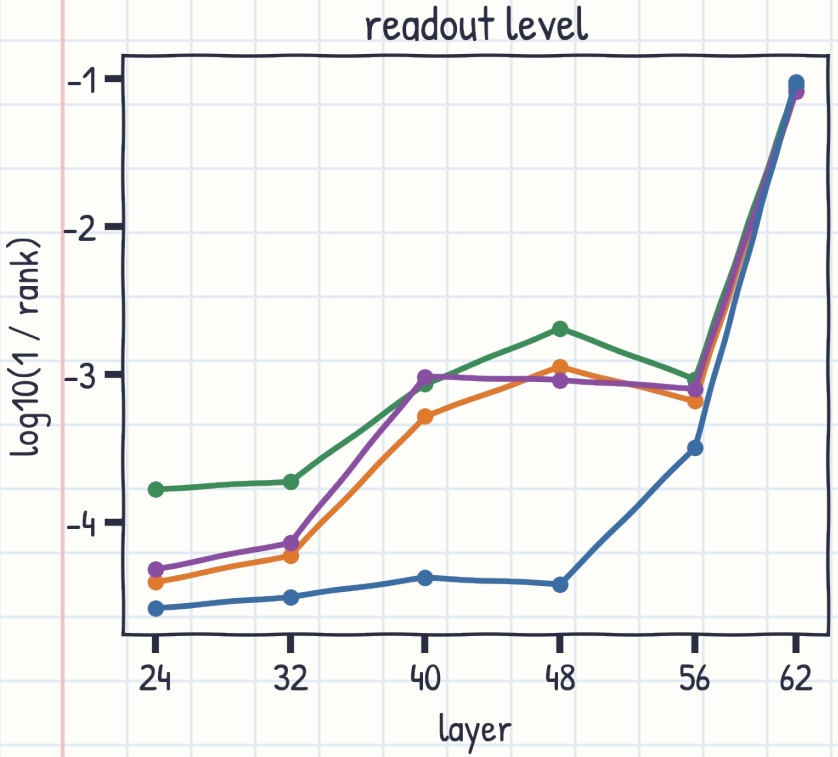
same-field paper: 0

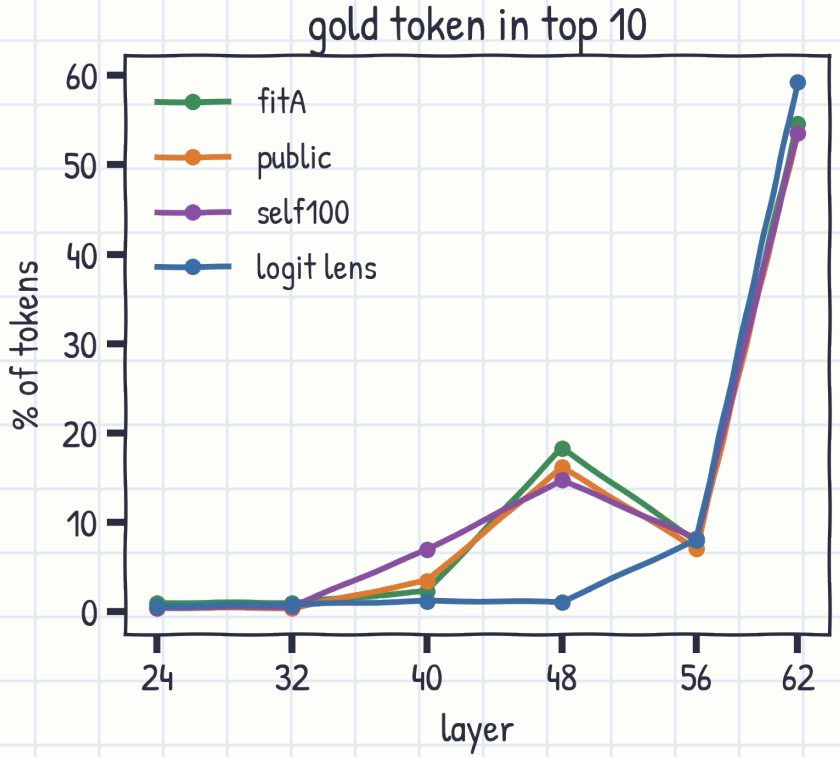
so it is this paper, already at L24

lenses cannot read it

Which lens reads best?

all papers · true title





Middle layers:

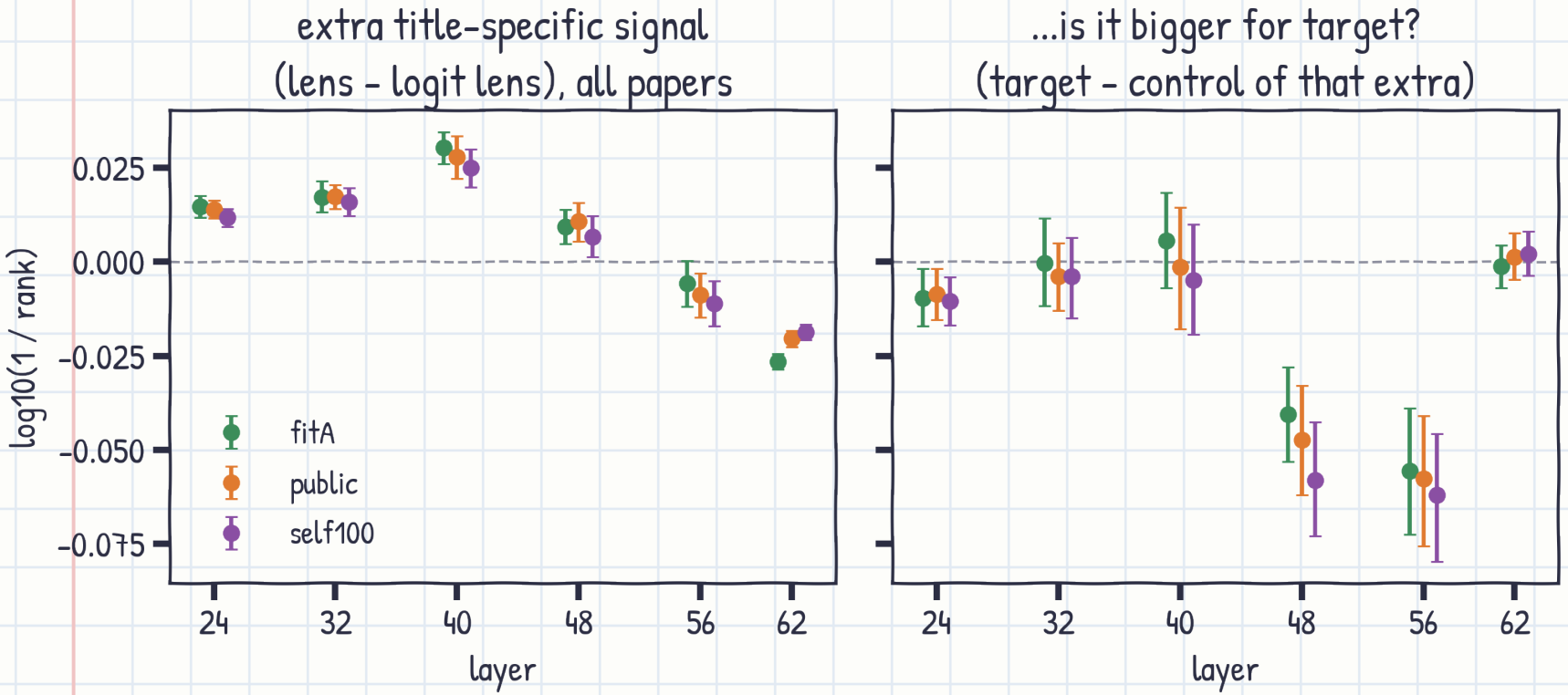
fitA best; at L48 15–18% top-10 vs 1% for logit

L62: logit lens best (59%)

Stage 10

Do lenses reveal hidden paper knowledge?

paired differences, 95% CI



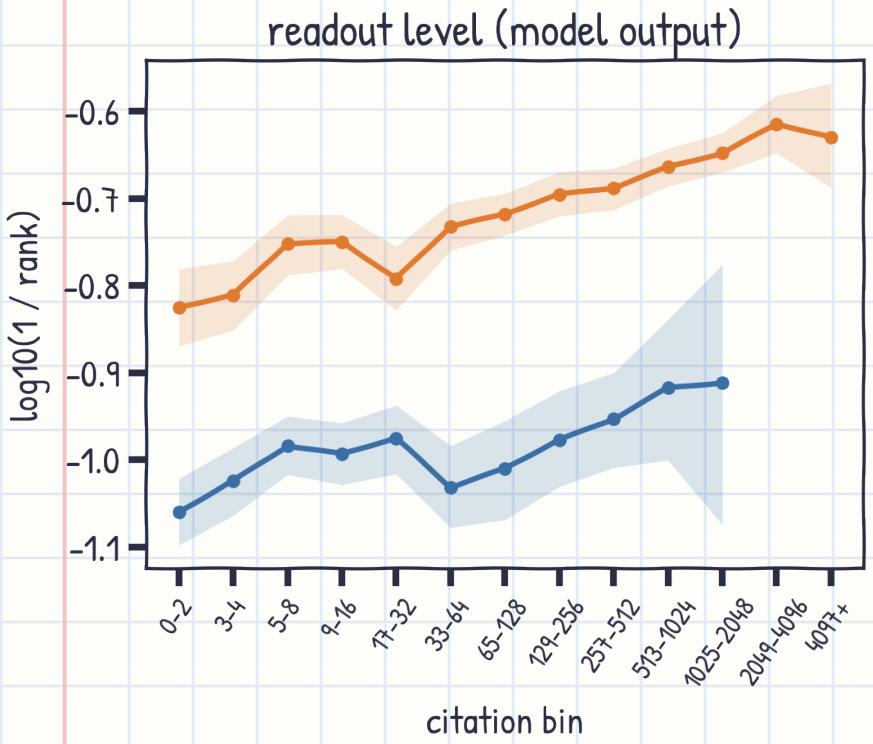
Extra signal: **+0.01 to +0.03**

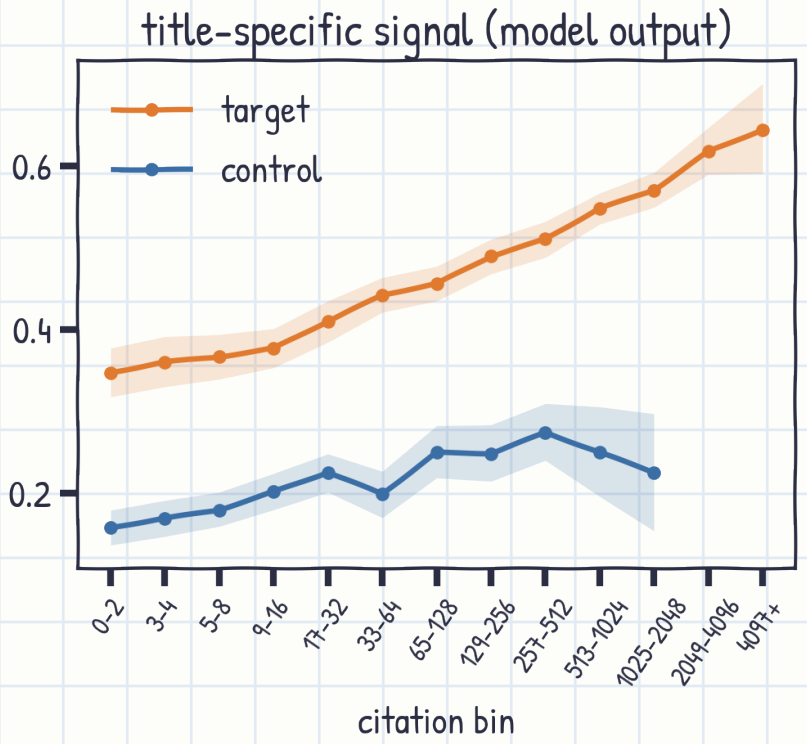
vs ~0.44 at L62

Not bigger for target
(even negative at L48–56)

More citations, more paper knowledge

model output · bins with < 10 papers omitted





Target: +0.35 -> +0.64

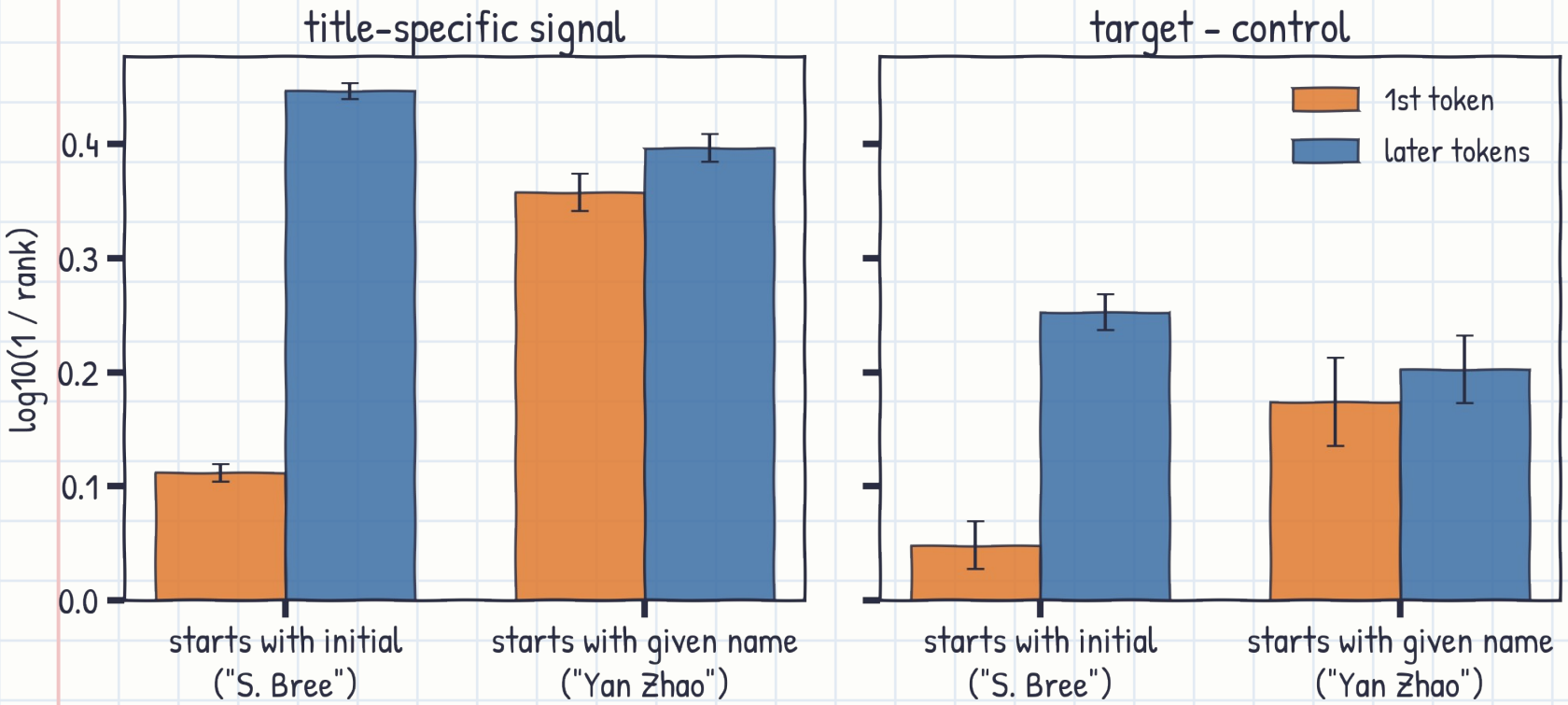
Spearman (title-specific)

0.45 all · 0.33 target · 0.19 control

Target > control in every bin

First token or the rest of the name?

model output · split by name format

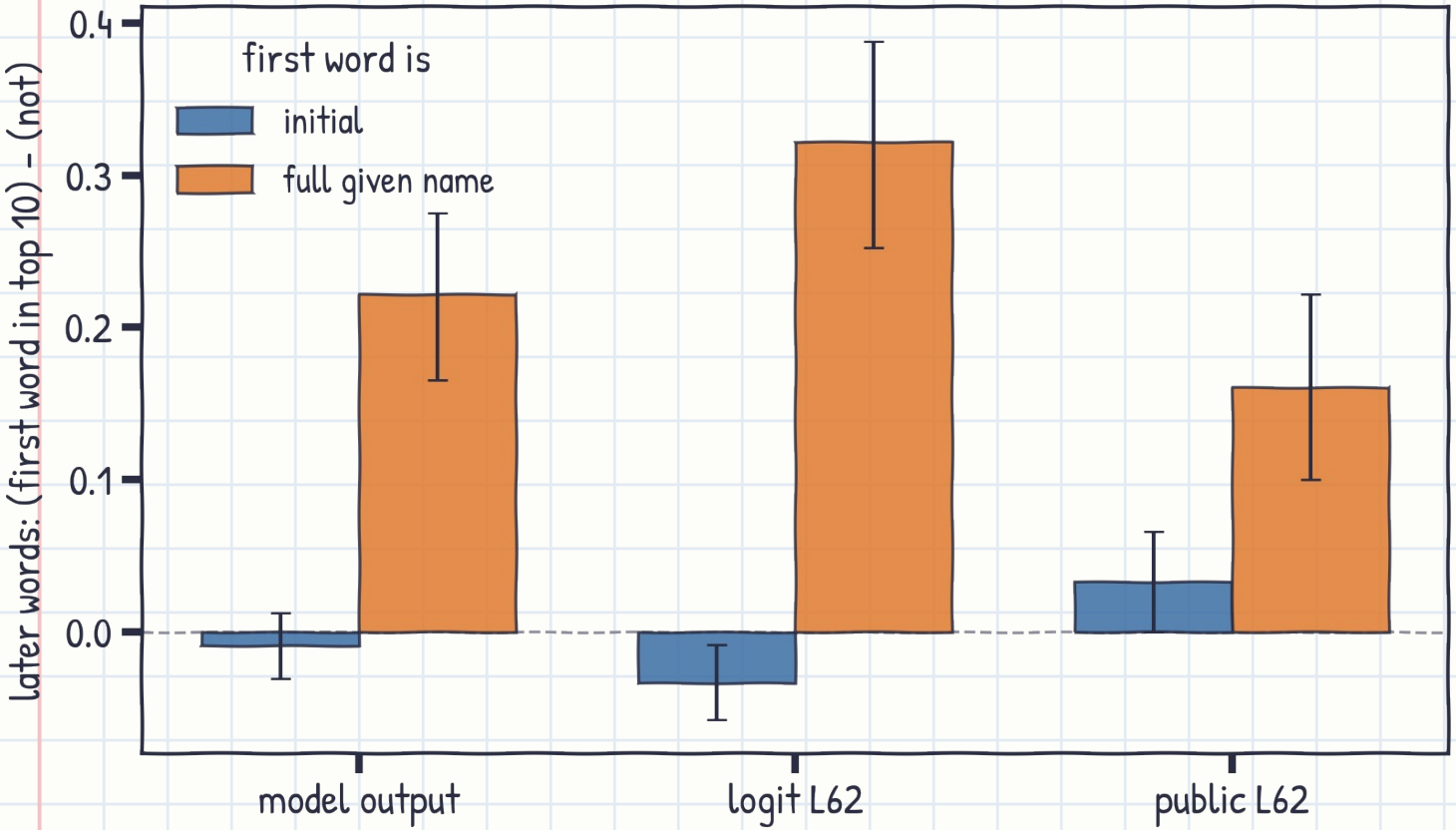


Initial: carries little; signal is on the surname

Full given name: carries about as much as the surname

Knowing the given name helps with the surname

true title · later words when the first word is / is not in the top 10



Given name in top 10

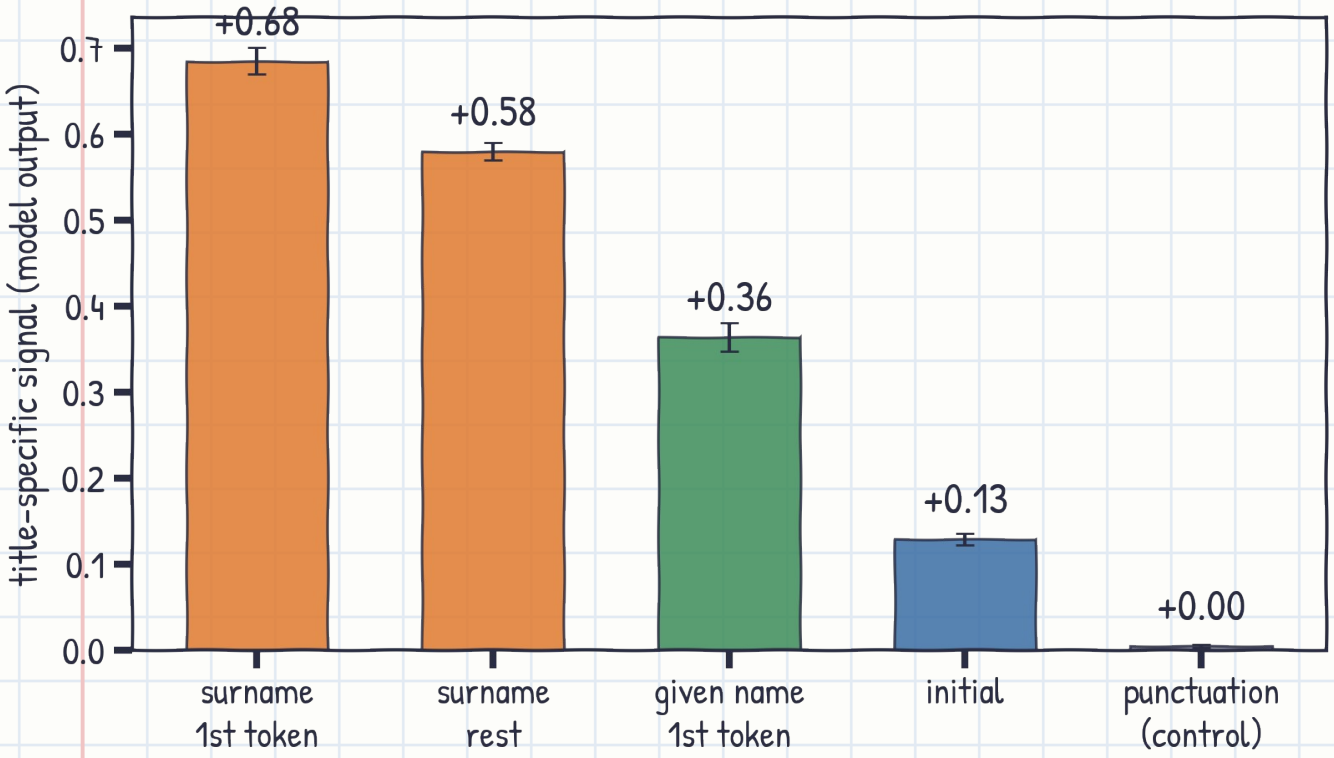
surname +0.22 (output)

+0.32 (logit L62)

Initial in top 10: no help

Where the signal sits, and a dose-response

model output





Right: R0/R1 papers, centred within citation bin; Spearman rho = 0.27, p < 0.001.